



# PHOTOSYNTHESIS IN HIGHER PLANTS

## INTRODUCTION

- Green plants synthesise their food by photosynthesis and all other organisms depend on plants for their needs.
- Photosynthesis is a physico-chemical process by which plants use light energy to drive the synthesis of organic compounds.
- The use of energy from sunlight by plants doing photosynthesis is the basis of life on Earth
- Photosynthesis is important due to two reasons : (a) It is the primary source of all food on earth and (b) It is also responsible for the release of oxygen into the atmosphere.

- Experiment for starch formation on variegated leaf or a leaf that was partially covered with black paper & exposed to light showed that photosynthesis occurred only in green part of leaves in the presence of light
- Experiment where a part of leaf is enclosed in a test-tube with some KOH soaked cotton (which absorbs CO<sub>2</sub>), while other half is exposed to air and set-up kept in light proved that CO<sub>2</sub> is needed for photosynthesis.

## WHAT DO WE KNOW?

## EARLY EXPERIMENTS

### 1) Joseph Priestley

Using a burning candle, a mouse, mint plant and a bell jar for closed space, hypothesised that plants restore the air whatever burning candles or breathing animals remove

### (2) Jan Ingenhousz

In an elegant experiment with an aquatic plant, showed that in bright sunlight plants produce oxygen.

### (3) Julius Von Sachs

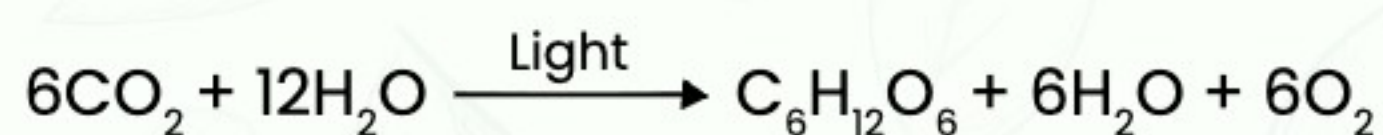
Found that glucose is made in green plant parts and stored as starch

### (4) T W. Engelmann

Using a prism, green alga *Cladophora* and aerobic bacteria, described the action spectrum of photosynthesis, which roughly resembles the absorption spectrum of chlorophyll- a and b.

### (5) Cornelius Van Niel

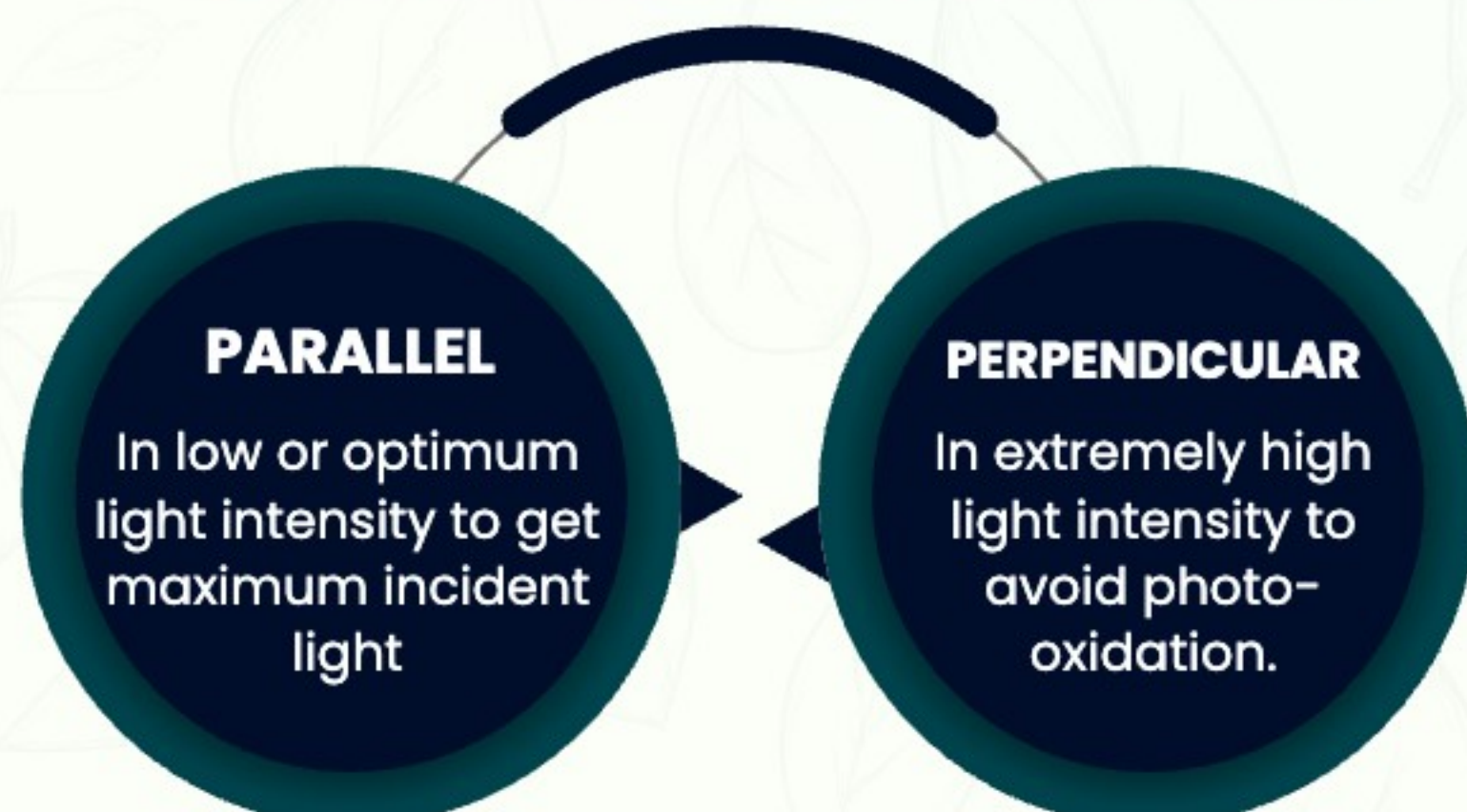
- Demonstrated that photosynthesis is essentially a light dependent reaction in which hydrogen from suitable oxidisable compound reduces CO<sub>2</sub> to carbohydrates.
- H<sub>2</sub>S is hydrogen donor for purple & green sulphur bacteria H<sub>2</sub>O the hydrogen donor in green plants which is oxidised to O<sub>2</sub>.
- The oxidation product is sulphur or sulphate in purple & green sulphur bacteria and not O<sub>2</sub>. Hence it was inferred that O<sub>2</sub> evolved by green plants comes from H<sub>2</sub>O and not from CO<sub>2</sub>. This was later proved by using radioisotopic techniques. The correct equation, for the overall process.



## WHERE DOES PHOTOSYNTHESIS TAKE PLACE

- In green parts of the plants, mainly in the mesophyll cells in the leaves, which have large number of chloroplasts.
- Usually the chloroplasts align themselves along the walls of mesophyll cells to get optimum quantity of the incident light.

## CHLOROPLAST ALIGNMENT



There is a clear division of labour within the chloroplast.

## CHLOROPLAST

### MEMBRANOUS SYSTEM

- (Grana + Stromal lamellae)
- Responsible for trapping light & synthesis of ATP & NADPH
- Directly light driven, called light reaction (photochemical reactions)

### STROMA

- Enzymatic reactions to synthesise sugar, which in turn forms starch, takes place
- Dependent on products of light reactions (ATP & NADPH)
- By convention called dark reactions (Carbon reactions)

However, this does not mean that the dark reaction occurs in darkness or that they are not light-dependent





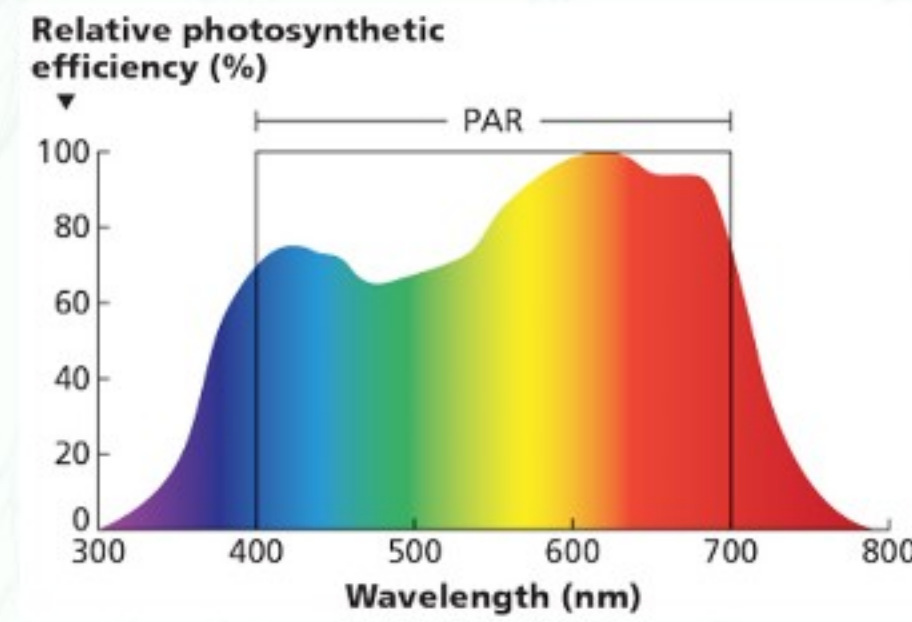
## HOW MANY TYPES OF PIGMENTS ARE INVOLVED IN PHOTOSYNTHESIS

Leaf-pigments of any green plant can be separated through paper chromatography

The colour in leaves is due to four pigments, that have the ability to absorb light, at specific wavelengths

### COLOUR OF THE PIGMENTS IN THE CHROMATOGRAPHY

- i. Chlorophyll-a = Bright or blue green
- ii. Chlorophyll-b = Yellow green
- iii. Xanthophyll = Yellow
- iv. Carotenoids = Yellow to yellow- orange



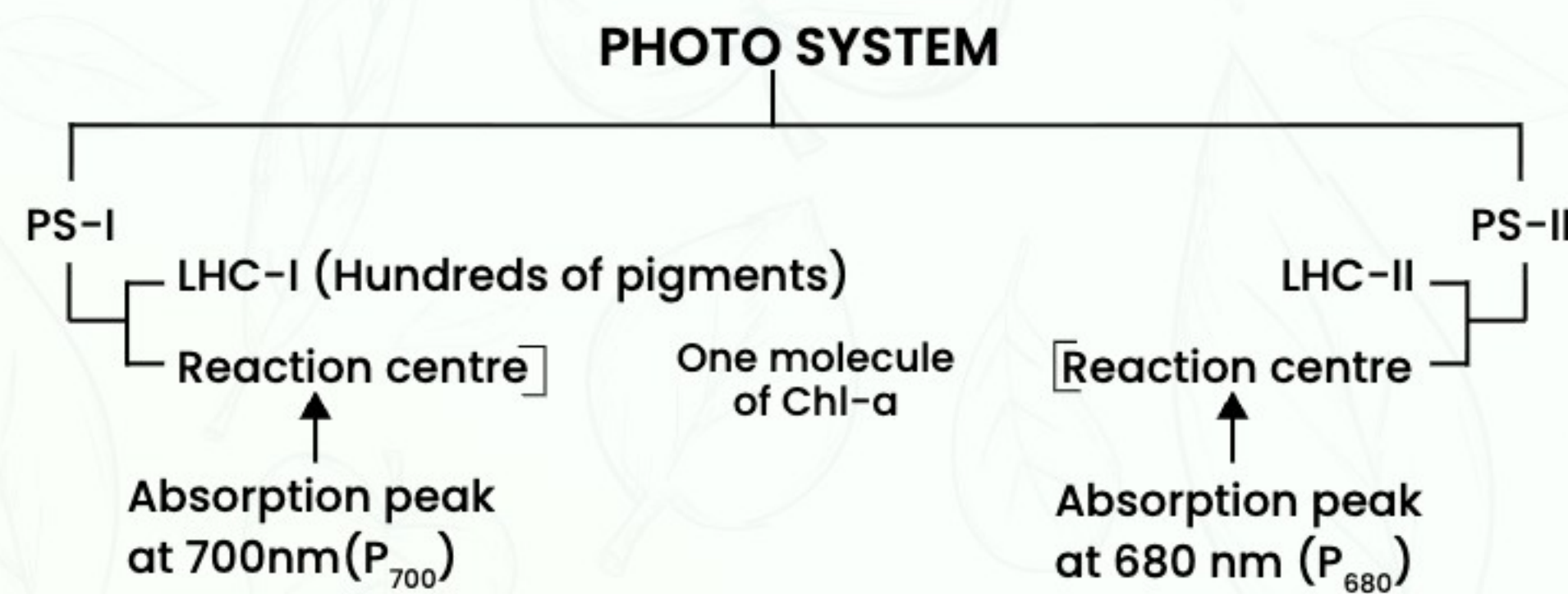
- The wavelength of light at which there is maximum absorption by chlorophyll-a i.e.. in blue and red regions, also shows higher rate of photosynthesis.
- Hence, we can conclude that Chl-a is the chief pigment associated with photosynthesis.
- Chl-b, carotenoids and xanthophyll are accessory pigments. They absorb light and transfer the energy to Chl-a. They enable a wider range of wavelength of incoming light to be utilised for photosynthesis and also protect chlorophyll-a from Photo-oxidation.

### WHAT IS LIGHT REACTION?

- Light reactions or the Photochemical phase include
  - (a) Light absorption (b) Water splitting
  - (c) Oxygen release, and (d) ATP and NADPH formation

- Several protein complexes are involved in the process

- The pigments are organised into two photosystems

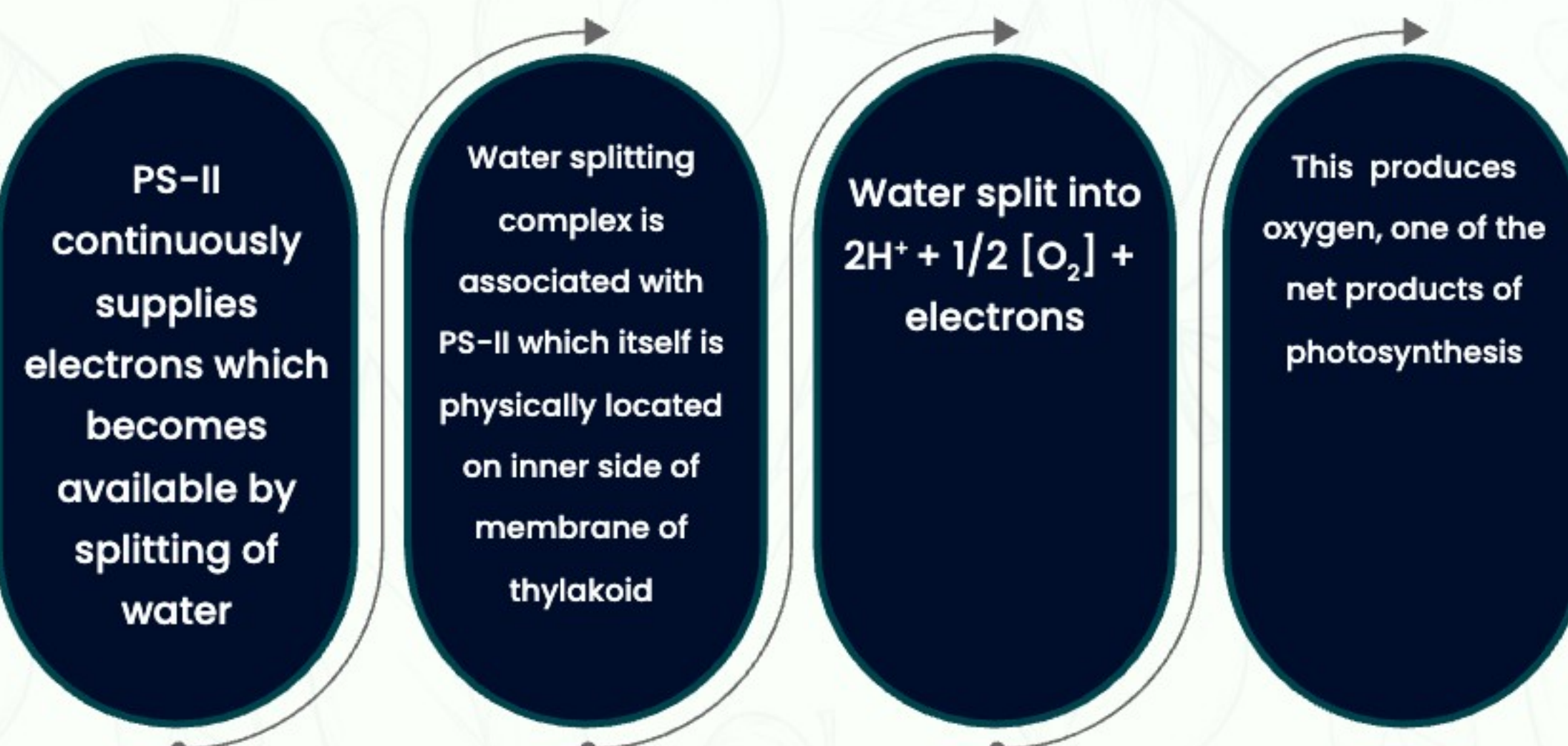


Named the sequence of their discovery and not in the sequence of their function.

### THE ELECTRON TRANSPORT

- The whole scheme of transfer of electrons starting from PS-II → uphill to the acceptor → down the ETC to PS-I → Excitation of electrons → transfer to another acceptor → finally downhill  $NADP^+ \rightarrow$  reducing it to  $NADPH \rightarrow H^+$  is called the Z-scheme due to its characteristic shape.
- This shape is formed when all the carriers are placed in a sequence on a redox potential scale.

### SPLITTING OF WATER



### CYCLIC AND NON-CYCLIC PHOTO-PHOSPHORYLATION

- When both PS-I and PS-II are involved the process is non-cyclic producing ATP,  $NADPH + H^+$  and oxygen
- When only PS-I is functional, cycle flow of electron takes place to produce only ATP.
- A possible location for cyclic flow is the stroma lamellae membranes which lack PS-II and NADP reductase enzyme.

Cyclic photo-phosphorylation also occurs when light only of wavelengths beyond 680 nm are available for excitation

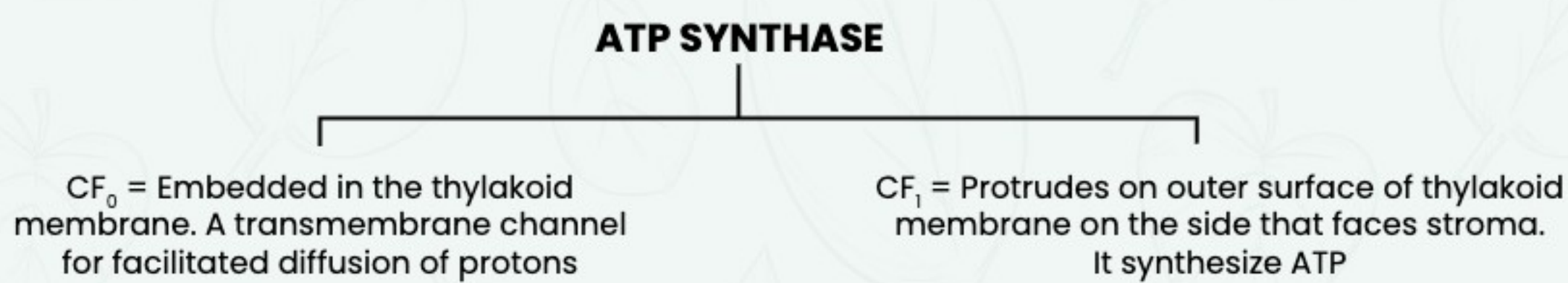
The membrane or lamellae of the grana have both PS I and PS-II



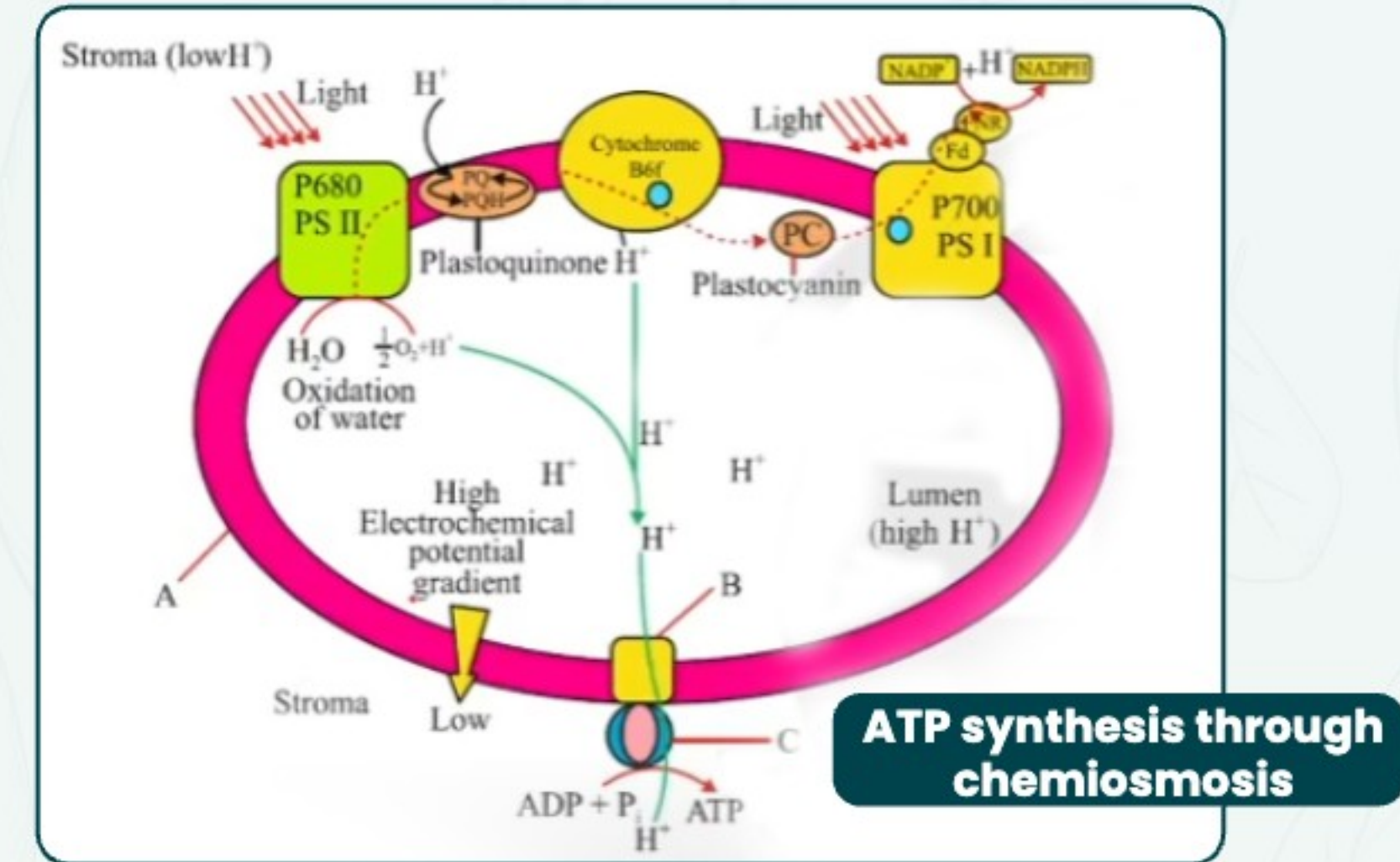


## CHEMIOSMOTIC HYPOTHESIS

- ATP synthesis during photosynthesis is linked to the development of a proton gradient across the membranes of thylakoid and protons accumulate in the lumen of thylakoids
- The proton gradient is caused by:
  - (a) Protons or hydrogen ions produced by splitting of water, accumulate in the lumen of the thylakoids.
  - (b) The primary acceptor of electron located towards outer side of membrane transfers its electron to a H carrier, which removes a proton from stroma while transporting an electron, to the electron carrier on the inner side of membrane
  - (c) The NADP reductase enzyme located on stroma side of membrane removes protons from stroma, while reducing  $\text{NADP}^+$  to  $\text{NADPH} + \text{H}^+$ .
- Within chloroplast, protons decrease in stroma and accumulate in lumen. This creates a proton-gradient across thylakoid membrane as well as a measurable decrease in pH in the lumen.
- Breakdown of this gradient leads to synthesis of ATP. When protons move across the membrane to the stroma through transmembrane channel of the  $\text{CF}_0$  of the ATP synthase.



- Chemiosmosis requires – a membrane, a proton pump, a proton gradient and ATP synthase



## WHERE ARE THE ATP AND NADPH USED?

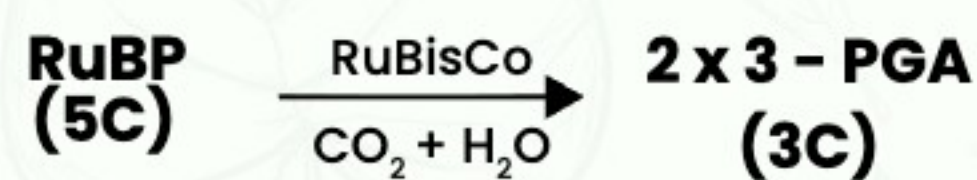
Of the products of light reaction– ATP, NADPH and  $\text{O}_2$ .  $\text{O}_2$  diffuses out of chloroplast while ATP and NADPH are used to synthesise sugars in the biosynthetic phase of photosynthesis. Melvin Calvin used radioactive  $^{14}\text{C}$  in algal photosynthesis studies to discover the first  $\text{CO}_2$  fixation product, the 3-C organic acid (3-PGA) ( $\text{C}_3$ - pathway).

In another group of plants, the first stable product was 4 carbon, oxaloacetic acid OAA ( $\text{C}_4$ -pathway)

## THE CALVIN CYCLE

- The Calvin cycle occurs in all photosynthetic plants whether they have  $\text{C}_3$  or  $\text{C}_4$  (or any other) pathways.
- Calvin cycle can be described under three stages.

**CARBOXYLATION:** Most crucial step



**REDUCTION:** A series of reactions that lead to formation of glucose. Utilises 2 ATP and 2 NADPH per  $\text{CO}_2$ . (The fixation of  $6\text{CO}_2$  and 6 turns of the cycle are needed to form one molecule of glucose from the pathway)

**REGENERATION:** Regeneration of RUBP is crucial for the cycle to continue. This step requires one ATP so, to produce one molecule of glucose in Calvin cycle an input of  $6\text{CO}_2$ , 18 ATP & 12 NADPH are required.

## THE $\text{C}_4$ -PATHWAY

Plants adapted to dry tropical regions have the  $\text{C}_4$ -pathway.

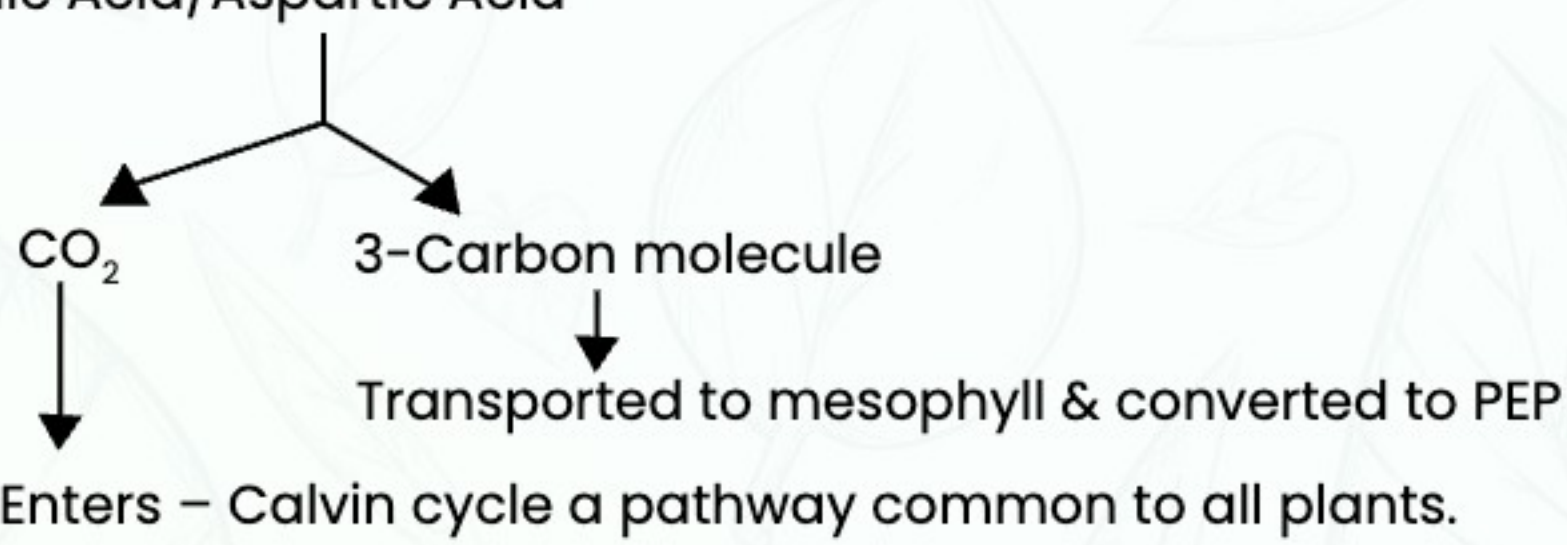
$\text{C}_4$ -plants are special: They have special type of leaf anatomy, tolerate higher temperatures, show response to high light intensities, lack photorespiration & have greater biomass productivity.

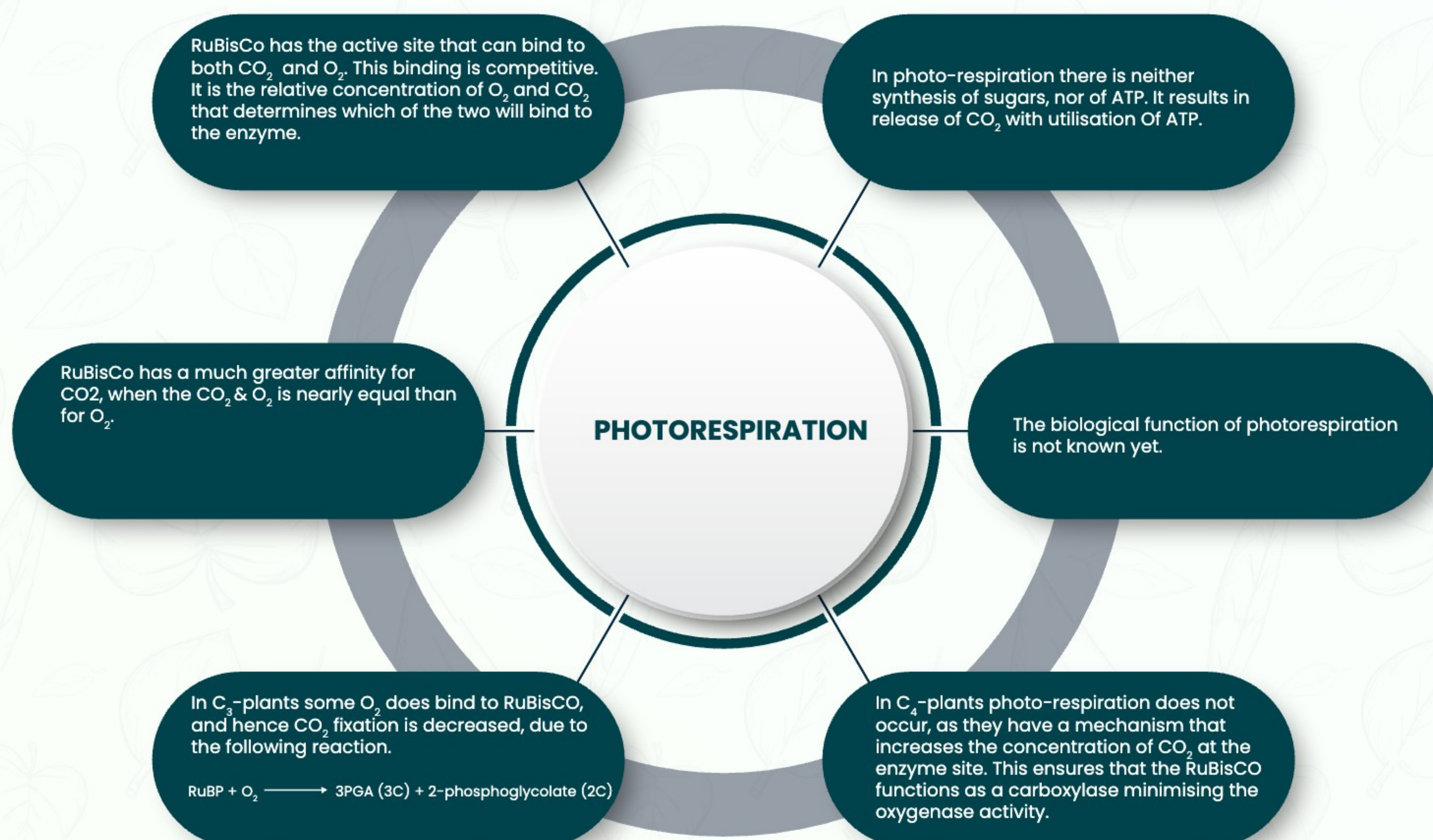
$\text{C}_4$ -plants have leaves showing Kranz anatomy, particularly have large cells around the vascular bundles, which may form several layers and are called bundle sheath cells, characterised by having a large number of chloroplasts, thick walls, impervious to gaseous exchange & no intercellular spaces.

The pathway is cyclic & called the Hatch and Slack Pathway. It is partly completed in mesophyll & partly in bundle sheath cell.



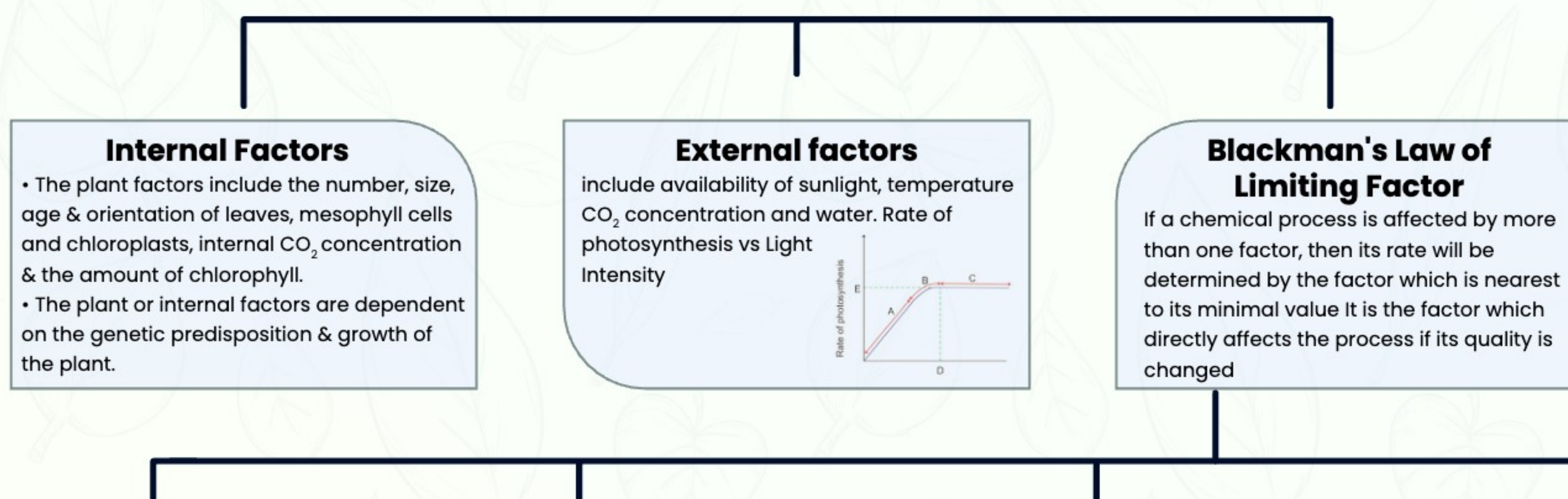


| MESOPHYLL CELL                                                                                                                                                                                                                                        | BUNDLE-SHEATH CELLS                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) Primary CO <sub>2</sub> acceptor is a 3-C compound PEP<br>(2) Enzyme for this fixation is PEPcase.<br>(3) Lacks RuBisCO<br>(4) C <sub>4</sub> -acid formed is OAA which forms malic acid or aspartic acid and transported to bundle sheath cells. | (1) Malic Acid/Aspartic Acid<br> (2) Rich in RuBisCO, but lack PEP case. |



## FACTORS AFFECTING PHOTOSYNTHESIS

Photosynthesis is under the influence of several factors both internal (plant) & external



**Light:** Light saturation occurs at 10% of the full sunlight except for plants in shade or in dense forests, light is rarely a limiting factor in nature

- There is a linear relationship between incident light & CO<sub>2</sub> fixation rates at low light intensities. At higher light intensities, gradually the rate does not show further increase as other factors become limiting.

**CO<sub>2</sub> concentration:** Major limiting factor. The concentration of CO<sub>2</sub> is very low in the atmosphere (0.03 & 0.04%). Increase in concentration upto 0.05% can cause increase in CO<sub>2</sub> fixation rates, beyond this levels can become damaging over longer periods.

- At low light conditions neither group (C<sub>3</sub> & C<sub>4</sub>) responds to high CO<sub>2</sub> conditions. C<sub>4</sub> plants show saturation at 360 μL<sup>-1</sup> CO<sub>2</sub> while in C<sub>3</sub>, saturation is seen at 450 μL<sup>-1</sup>. Some greenhouse crops like tomatoes and bell pepper show higher yields in CO<sub>2</sub> enriched atmosphere.

**Temperature:** Dark reactions being enzymatic are temperature controlled, Light reactions are also temperature sensitive but to much lesser extent

C<sub>4</sub>-plants show higher yield at high temperature while C<sub>3</sub>-plants have a much lower temperature optimum.

**Water:** Effect of water as a factor is more through its effect on the plant rather than directly on photosynthesis. Water stress causes the stomata to close hence reducing CO<sub>2</sub> availability. Water stress also makes leaves wilt, thus, reducing the surface area of leaves and their metabolic activity as well.